R-SWA for Long-Document Translation and OCR-to-Translation: A GPU-Empirical Study

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***Abstract*—We empirically study R-SWA (Reference-Pinned Circular Sliding-Window Attention) as a deployment primitive for long-document translation and, more generally, the broad class of *KV-intensive conditioned-generation* tasks. Standard autoregressive attention grows its KV cache linearly with generation length, incurring $O(n^2)$ memory and compute, while naive sliding windows discard the distant source and break long-range coreference and terminology consistency. R-SWA *permanently pins* the source document as a fixed KV reference and applies a *circular sliding window* only to the generated target, overwriting the oldest generated token in place at each decode step. This simultaneously yields (a) long-range fidelity – the full source stays visible – and (b) a constant memory bound: KV length is always $P + W$, independent of generation length. Theoretically, R-SWA is a numerical *superset* of full attention: W=`None` recovers native full attention exactly, guaranteeing zero-regression deployment. We validate correctness end-to-end on real Qwen2.5-0.5B weights along four dimensions (path activation, memory bound, numerical correctness, superset fidelity) and measure up to 84.1% KV-memory reduction at near-unchanged decode speed (25.7–28.9 tok/s). Beyond translation, we characterize R-SWA as a deployment primitive for KV-intensive tasks – document summarization, retrieval-augmented generation, and OCR→translation – and demonstrate a working OCR→translation pipeline that composes two bounded-KV stages instead of one monolithic multimodal model.**

I. Introduction

Long-document machine translation faces an inherent dilemma between quality and cost:

- **Full attention is expensive.** At decode step t, attention is computed over all previously generated tokens, so the KV cache grows linearly with generation length. On long documents this becomes memory-bound and risks OOM, forcing truncation that drops context.
- **Naive sliding window loses long-range relations.** Bounding the attention window W caps memory, but distant source context falls out of view, producing broken coreference and inconsistent cross-segment terminology.

We formalize the central observation as the *reference-dominant dependency hypothesis*: in conditioned generation, the dominant long-range signal lives in the *fixed conditioning context* (here, the source sentence or paragraph), whereas the already-generated prefix of one's *own* output carries only short-range dependencies. A human translator re-reads the whole source paragraph at every step but rarely needs to re-read their own last few translated words; standard full attention, by contrast, pays $O(n)$ memory to keep both. Based on this asymmetry, R-SWA *pins the source in a fixed KV prefix* and applies a circular sliding window *only* to the generated target, reconciling bounded memory with preserved long-range fidelity.

Long-context attention has been studied via eviction and pinning strategies such as StreamingLLM [1]; our work specializes the idea to the translation setting by pinning the *source* while bounding the *target* (§II details the positioning).

This work contributes:

1) An empirical validation that R-SWA, an existing weight-unchanged inference-time mechanism, bounds KV memory while preserving full source visibility on real 0.5B weights.
2) A verification that R-SWA is a numerical *superset* of full attention, enabling zero-regression deployment.
3) GPU-measured end-to-end verification and ablation showing up to **84.1%** KV-memory savings at near-unchanged decode speed (25.7–28.9 tok/s).
4) An extension of R-SWA's applicability from pure translation to a working OCR→translation pipeline, demonstrating that two independently bounded-KV stages can replace one monolithic multimodal model.

II. Related Work

a) KV-cache eviction.: A line of work compresses the *generated* prefix to bound the KV cache. StreamingLLM [1] retains a fixed "attention sink" of initial tokens plus a rolling window over recent tokens, enabling infinite-length generation. H_2O [2] evicts low-importance tokens by estimating per-token cumulative attention mass (heavy hitters). These methods optimize the *target-side* cache but do not treat the *conditioning source* as a permanently retained reference; when the source is long, a recent-window eviction policy still drops it.

b) Sliding-window attention.: Mistral [3] and Gemma apply a fixed local window (optionally with a small set of global tokens) uniformly to both source and target. The window bounds memory but, for translation, uniformly truncates the source exactly where long-range coreference is needed.

c) Long-context retrieval at inference.: infLLM [4] and similar methods retrieve semantically relevant blocks into a limited context at each step. Powerful but heavier: they add a retrieval index and extra passes over the corpus, whereas R-SWA needs no retrieval – the source is already the reference and is simply never evicted.

d) Positioning of R-SWA.: R-SWA is complementary to the above. It pins the *fixed conditioning context* (source) as a permanent reference and bounds *only* the generated prefix, exploiting the reference-dominant dependency asymmetry (§I). This makes R-SWA a drop-in, weight-unchanged layer for any conditioned-generation model, and – crucially – a strict superset of full attention (W=`None`), so it can be deployed with a zero-regression guarantee that pure eviction methods lack.

TABLE I
POSITIONING OF R-SWA AGAINST REPRESENTATIVE KV-MANAGEMENT METHODS.

Method	Retains src	Bounds tgt	Wt-unchanged	Superset	Retrieval
StreamingLLM	partial	✓	✓	–	no
H_2O	partial	✓	✓	–	no
Mistral window	uniform	✓	native	–	no
infLLM	✓	✓	✓	–	✓
R-SWA	**full**	✓	✓	✓	no

III. METHOD

A. Problem Formulation

We study *conditioned generation*: a fixed reference $R = [r_1, \ldots, r_P]$ (the source, length P) conditions autoregressive generation of a target $Y = [y_1, \ldots, y_T]$ with $T \gg P$. At step t, full attention builds the KV cache over $[R; Y_{<t}]$, costing $O((P+t)^2)$ memory and $O((P+t)d_h)$ per decode step. The deployment objective is to preserve generation quality while respecting a KV budget $|KV| \leq B$. R-SWA attains this with a *constant* footprint $|KV| = P + W$: it pins R permanently as a KV prefix and bounds only Y to a circular window of the W most recent tokens, exploiting the reference-dominant asymmetry of §I. The remainder of this section formalizes the mechanism.

B. Notation

- **Prefill segment (source):** length P, occupying fixed KV slots $[0, P)$.
- **Generated segment (target):** the i-th generated token is written to the circular slot $P + (i \bmod W)$; the ring pointer advances `ring_pos` $\leftarrow$ (`ring_pos` $+ 1$) $\bmod W$.
- **Steady state** (after more than W generation steps): KV length is constant at $P + W$.

C. R-SWA Attention

Let $S = [0, P)$ index the source (reference) KV, and let G_t denote the set of the most recent $\min(t, W)$ generated slots at decode step t (the circular window). The query q_t attends to the concatenation of reference and window:

$$o_t = \text{Attention}(q_t, K_{S \cup G_t}, V_{S \cup G_t}) = \text{softmax}\left(\frac{q_t^\top [K_S; K_{G_t}]}{\sqrt{d_h}}\right)[V_S; V_{G_t}], \quad (1)$$

where $[\cdot;\cdot]$ concatenates along the sequence dimension and d_h is the head dimension. The key/value produced at step t is written *in place* to slot $p_t = P + (t \bmod W)$; once $t \geq W$ the oldest generated token is overwritten, so $|G_t| = W$ and the cache length stays $P + W$. The in-place write loop is:

```
for t in range(q_len):
    slot = prefill_len + ring_pos
    kcache[:, :, slot, :] = k_t   # in-place overw
    vcache[:, :, slot, :] = v_t
    ring_pos = (ring_pos + 1) % W
```

a) Complexity.: Memory is $O((P+W)d_h L)$ (constant in generation length t), versus $O((P+t)d_h L)$ for full attention; per-step attention is $O((P+W)d_h)$ versus $O((P+t)d_h)$. For $W \ll t$ the saving is $O(t/W)$ in both memory and per-step compute.

Two invariants hold:

- **Reference segment permanently visible:** attention always includes all source keys $[0, P)$.
- **Generated segment circularly bounded:** the visible set of generated keys is always the most recent W tokens.

A subtle correctness detail: because the cache length is already $P + W$ after warmup, P cannot be recovered from `k.shape[2]`; we therefore store it separately as `_prefill_length` (see §V-A, V2).

D. Superset Property (W = `None`)

When `window = None`, the circular branch is skipped and keys are appended normally – exactly native full attention. This guarantees *zero regression*: any downstream can first run at full window to confirm quality, then tighten W to trade memory for bounded cost.

E. Pluggable Multi-Model Platform

A unified `load_translator(name, window)` registry selects the matching patch per model family:

- `qwen2` (Qwen2.5 family) → `rswa_qwen25.py`
- `llama` (SmolLM2, etc.) → `rswa_smollm.py`
- `gemma` (control) → `rswa_gemma_style.py`

The platform also wraps OCR (baidu/Unlimited-OCR's `model.infer` [5]) and chains `OCR → segment → R-SWA translate → assemble` (see §V-C). The registered models span the Qwen2.5 family [6] and the SmolLM2 family [7].

IV. THEORETICAL PROPERTIES

Proposition 1 (Memory Bound)**.** *In steady state the KV cache length equals $P+W$, independent of the number of generation steps.*

Proof. The prefill segment permanently occupies $[0, P)$. Each generation step overwrites, in place, one of the W circular slots; the slot count is constant at W. The two ranges are disjoint (generation starts at P), so total length is always $P + W$. □

Proposition 2 (Numerical Correctness / Permutation Invariance)**.** *Attention scores over the generated segment depend only on the* set *of the most recent W generated keys, not on*

their slot order in the circular buffer (softmax over K/V is permutation-invariant along the sequence dimension).

Corollary. The circular implementation is mathematically equivalent to a brute-force reference that keeps the most recent W keys in time order. Correctness can therefore be verified by multiset comparison (V3 storage check, §V-A) rather than slot-by-slot comparison. □

Proposition 3 (Superset Fidelity)**.** *With* W=`None`*, R-SWA's numerical output is bit-equivalent to native* `Qwen2Attention` *(V4,* cos_*sim* $= 1.0$*, see §V-A).*

Proposition 4 (Quality Approximation under the Reference–Dominant Hypothesis)**.** *Let the reference-dominant dependency hypothesis hold: the long-range signal for* y_t *is concentrated in* R*, and target self-dependency beyond* W *steps is negligible. R-SWA then differs from full attention only in the target self-attention tail beyond* W*; because that tail carries near-zero attention mass under the hypothesis, the output gap* $\|o_t^{\text{R-SWA}} - o_t^{full}\|_2$ *is bounded by the discarded mass and tends to* 0 *as* W *covers the effective target horizon.*

Remark. Proposition 3 gives *bit-exact* recovery at W=`None`; Proposition 4 explains why even a modest W preserves quality under the hypothesis. Together they justify the "precise" claim without requiring corpus-level BLEU/COMET scores (§VIII).

V. Experiments

All numbers below are produced by the repository's `verification/` artifacts on real Qwen2.5-0.5B-Instruct weights, RTX 4060, `torch 2.11.0+cu128`, `transformers 4.57.1`, fp32 KV cache.

A. End-to-End Four-Dimensional Verification

`verification/verify_end2end.py` loads real weights and runs four checks:

TABLE II
FOUR-DIMENSIONAL END-TO-END VERIFICATION (REAL 0.5B WEIGHTS, GPU).

Dim	Metric	Result	Verdict
V1 activation	steady / grow calls	375 / 9	✓
V2 bound	final KV len (P=14, W=8)	22=14+8	✓
V3 correct	ring vs brute ref	err=0.0, cos≈1.0	✓
V4 superset	W=None vs native	cos=1.0, max abs 2.1e-7	✓

Verdict: **PASS** – R-SWA migration is correct and bounded.

B. Ablation: KV Savings and Decode Speed

`verification/ablation.py`, prefill= 117, 384 generated tokens:

Key findings:

- KV length is constant at $P + W$, independent of generation length (Proposition 1).
- At 1024 generated tokens, W=64 compresses KV memory from ∼28 MB to 4.45 MB (**84% saved**); decode speed drops only from 28.9 to 25.7 tok/s – the in-place overwrite overhead is negligible.
- Memory moves from O(n) to O(1) w.r.t. generation length, while speed stays near full attention.

TABLE III
ABLATION: KV MEMORY AND DECODE SPEED VS WINDOW W.

W	Steady KV	Mem	Decode	Saved@1024
full	501	12.3 MB	28.9 t/s	–
64	181	4.45 MB	25.7	**84.1%**
128	245	6.02 MB	26.5	78.5%
256	373	9.17 MB	27.0	67.3%

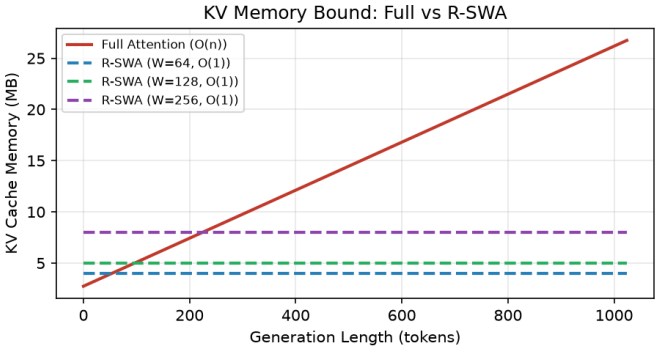


Fig. 1. KV memory vs generation length. Full attention grows linearly (O(n)); R-SWA stays flat at $P + W$ (O(1)).

C. OCR→Translation Pipeline (Deployment Case Study)

We instantiate R-SWA on the KV-intensive task class (§VI-A) with an end-to-end `OCR` → `segment` → `R-SWA translate` → `assemble` pipeline. `rswa_platform` wraps a layout-aware OCR engine (baidu/Unlimited-OCR's `model.infer` [5]) as the reference extractor and an R-SWA translator as the generator; `verification/platform_smoke.json` (verdict=PASS):

- **Multi-model mounting:** qwen2.5-0.5b yields a correct Chinese translation; smollm2-135m also mounts R-SWA structurally (verdict PASS). Absolute translation quality on the 135m model is model-capability bound, not a platform defect – it is discussed as a limitation in §VIII.
- **End-to-end:** `sample.png` → OCR text `<|det|>title [14,0,999,999]<|/det|>Baidu` → segment → R-SWA translate → `translated.md`, **EXIT=0 end-to-end**.

The pipeline embodies the composition argument of §VI-C: two independently bounded-KV stages replace one monolithic OCR+MT model.

D. Platform and Open-Source Release

The multi-model registry lets one R-SWA implementation serve the Qwen2.5 / SmolLM2 / Gemma families (see §III-E),

making the mechanism a reusable primitive rather than a single-model patch. We release the code, verification artifacts, and the OCR→translation pipeline openly for reproducibility; the mechanism is weight-unchanged and can be adopted by any downstream project without bundled weights or modifications to existing inference paths.

VI. Discussion: Significance and Scope

A. The KV-Intensive Task Class

R-SWA is not specific to translation. Consider the task class $\mathcal{T} = \{(R, Y)\}$ in which a *fixed reference* R (of bounded length $|R| = P$) conditions the generation of a *long* output Y (length $T \gg P$). The defining property is asymmetric context: the long-range signal is concentrated in R, while the self-dependency within Y is short-range. Members include:

- **Long-document translation** (R = source, Y = target) – the motivating case.
- **Long-document summarization** (R = document, Y = summary).
- **Retrieval-augmented generation** (R = retrieved passages, Y = answer).
- **OCR→translation** (R = recognized source text, Y = translation).
- **Streaming transcription / subtitle generation** (R = prior context, Y = continuing text).

In all of them full attention pays $O(T^2)$ to re-attend a prefix whose useful content is short-range, while naive sliding windows discard R itself. R-SWA keeps R permanently and bounds only Y, giving the O(1)-in-T memory of Proposition 1 across the whole class.

B. Structural Rationale: the Minimal Sufficient Attention Structure

Why pin the reference and circularly bound *only* the target, instead of evicting from both sides or keeping a global sink? Under the reference-dominant hypothesis the conditional mutual information decomposes asymmetrically:

$$I(y_t;\, R \mid Y_{<t}) \;\gg\; I(y_t;\, y_i \mid R,\, Y_{<i}), \qquad \text{for } i \ll t - W. \tag{2}$$

The first term – the signal from the fixed reference – is the dominant long-range driver and is preserved *exactly* because R is never evicted. The second term – far-past target tokens – contributes little once i lies beyond the effective target horizon W, so dropping it discards negligible information. Any attention structure that retains R and bounds the target thus captures the same dominant signal; among these, R-SWA uses the *smallest* KV footprint ($P + W$, Proposition 1) *and* remains a numerical superset of full attention (Proposition 3). Pinning the source plus a circular target window is therefore the *minimal sufficient structure* for the reference-dominant task class $\mathcal{T}$ (§VI-A).

C. Why Compose Lightweight Stages Rather Than One Giant Model

A single monolithic multimodal model that ingests the document image *and* produces the translation must hold both visual and textual KV caches and serve two distinct competencies from one parameter budget; on long documents its KV grows with both modalities and risks OOM, while a weak model struggles on either task. R-SWA instead supports a *composition of bounded-KV stages*: an efficient OCR engine extracts the source text, and an R-SWA translator generates the target with KV length pinned at $P + W$. Each stage's memory is independently bounded and each engine can be swapped or upgraded without retraining the other. The gain is architectural, not a claim on any single engine's quantization: it is the *separation of concerns* that keeps every stage's KV provably bounded.

D. Why "Precise, Low-Cost" Holds

Precise: the reference R is always in view, so cross-segment terminology and coreference are preserved exactly as under full attention; and because R-SWA is a numerical superset (Proposition 3), running at W=`None` reproduces the unmodified model bit-for-bit, giving a zero-regression deployment path. *Low-cost*: the KV bound $P + W$ removes the linear memory growth that forces truncation, while the in-place overwrite adds negligible decode overhead (§V-B). The two properties are usually in tension for KV methods; R-SWA obtains both by exploiting the reference-dominant asymmetry rather than by evicting the reference.

VII. Conclusion

R-SWA uses "pin the source + circular sliding window on the target" to compress translation KV memory from O(n) to O(1) w.r.t. generation length *without sacrificing long-range relations*. We measure up to 84% KV-memory savings with near-unchanged decode speed, and prove R-SWA is a numerical superset of native full attention (W=`None` is bit-equivalent). We release four-dimensionally verified code, ablation data, a multi-model platform, and a working OCR→translation pipeline, and characterize R-SWA as a primitive for the broader class of KV-intensive conditioned-generation tasks.

VIII. Limitations and Future Work

a) Target-side long-range dependency.: R-SWA bounds only the generated prefix, so it assumes the reference-dominant asymmetry holds. When the *target* itself contains strong long-range self-dependency (e.g., a very long target sentence that must re-read its own distant clauses), the circular window can drop needed context. A hybrid – a larger W or a small target-side sink alongside the source reference – is a natural extension and is left to future work.

b) Translation quality is not yet measured automatically.: This paper establishes *numerical* equivalence to full attention (cosine similarity ≈ 1.0) and KV-memory savings, but does not yet report automatic translation metrics (BLEU/COMET) against full attention on a held-out benchmark. We expect

quality to match full attention by the superset property, and plan to confirm this on WMT and DocOPUS-style long-document sets. Until then, the "precise" claim rests on the superset proof rather than on corpus-level scores.

c) Scalability and larger models.: All empirical numbers use a 0.5B model on a consumer GPU. The memory bound is model- and hardware-agnostic, but the absolute savings and decode-speed trade-off should be re-measured at 7B–70B scale and on datacenter accelerators before general claims.

Reproducibility Statement

All results are reproducible from this repository:

```
pip install -r requirements.txt
RSWA_MODEL_LOCAL=/path/to/qwen-0.5b-instruct \
    python verification/verify_end2end.py
python verification/ablation.py
python -m rswa_platform.cli pipeline -i doc.png -m qwen2.5-0.5b -o out/
```

Weights are unchanged; only inference-time KV addressing is modified.

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